Certified Personal Trainer

DOMAIN FOUR

PROGRAM DESIGN AND IMPLEMENTATION

EXAM WEIGHT 25%

ISSA INTERNATIONAL SPORTS SCIENCES ASSOCIATION

Before designing an exercise program, a trainer must gather information from the client's assessments. The trainer should take time to review the assessment results within the context of the intake paperwork, make comparisons to established norms and standards, and determine what the collective data means for the health of the client. Assessment data should then be used to design a customized program with the goal of keeping clients safe, moving effectively, and working toward their unique fitness goals.

The type of training program that will deliver the best client results is one that's comprehensive, individualized, periodized, and progressive. Program design does not need to be complicated or confusing. It should be an organized blueprint for solving the client's problems and helping them achieve their goals. A trainer will do this by using their knowledge of key training principles and acute program variables to influence physiological adaptations.

Elements of Fitness - Chapter 8

There are several different elements of a fitness program. Not all the elements need to be addressed in every workout, but they should all be considered at some point, to some degree, in a comprehensive fitness program.

- The warm-up
  - Necessary for preparing the body for the activity or training; purposes include:
    - Increase blood flow
    - Increase respiration
    - Increase body temperature
    - Increase neurological activation
  - Different types
    - General warm-up: nonspecific, low-intensity activity including dynamic stretching and light cardiovascular activity with the purpose of increasing blood flow, respiration, and body temperature
    - Specific warm-up: movements used to prepare the body for a sport or specific exercises
      - Variations of dynamic stretching and light cardiovascular activity
- Flexibility training
  - Element of fitness using stretching to increase the range of motion of a joint or group of joints
  - Flexibility and range of motion typically begin to diminish at age 25
  - Should be incorporated a minimum of three days a week
  - Different types
    - Static stretching: lengthening a muscle and holding the lengthened position
      - Ideal after exercise or on rest days
    - Dynamic stretching: movement-based active stretching where muscles engage to bring about a stretch
      - Ideal before exercise
        - May reduce muscle soreness

- Self-myofascial release (SMR): applying manual pressure to an adhesion or overactive tissue to elicit an autogenic inhibitory response, which is characterized by a decrease in the excitability of a contracting or stretched muscle arising from the Golgi tendon organ
  - Ideal before and after exercise
  - Example: foam rolling
  - Avoid the lumbar spine
- Active stretching: a type of stretching that involves a muscle actively contracting to stretch another muscle
- Passive stretching: a type of stretching that uses an external force to move a joint to the end of a range of motion
- Pre-contraction stretching: flexibility technique that may be assisted by or utilizes a stretching tool like a towel or stretching strap

▶ Flexibility training progression
- General warm-up
- Dynamic stretching, SMR
- Specific warm-up
- Exercise bout
- Static/passive or pre-contraction stretching, SMR

▶ Benefits of flexibility training
- Better posture
- Improved balance
- Decreased chronic pain
- Improved ROM

- Core training

▶ Core musculature:

Hip adductors	Gluteus medius	Gluteus minimus
Erector spinae	Rectus abdominus	Gluteus maximus
Hamstring complex	Quadriceps	Hip flexors
Transverse abdominus	Internal obliques	External obliques
Multifidus	Pelvic floor muscles	

- Abdominal muscles aren't the only muscles included in the core, so it involves more than targeted abdominal work
- Abdominal bracing works to activate the muscles of the trunk to support the spine and hold the pelvis in a neutral position
- Necessary for every client; research suggests that a strong core contributes to increased sports performance, stability, reduced back pain, and increased functional strength for everyday activity
- Example exercises
  - Abdominal crunch
  - Plank
  - Cable woodchop

- Balance training
  - Requires sensory input from the eyes, ears, and proprioceptors
  - Benefits
    - Improved static and dynamic stability
    - Reduced incidence of a recurrent ankle injury
    - Reduced low back pain (with core training)
    - Reduced joint pain (with strength training)
  - Should become progressively more difficult
    - Floor: two legs
    - Floor: single-leg with step
    - Floor: single-leg with ball
    - Floor: single-leg
    - Balance pad: two legs
    - Balance pad: single-leg with step
    - Balance pad: single-leg with ball
    - Balance pad: single-leg
- Reactive training
  - Although sports are most often associated with the need for speed, agility, and quickness (SAQ), nonathletes can benefit from the incorporation of reactive training into their fitness programs as well
    - Speed: the ability to move the body in one direction as fast as possible
    - Agility: the ability to accelerate, decelerate, stabilize, and change direction with proper posture
    - Quickness: the ability to react and change body position with a maximum rate of force production

- Plyometric training: reactive training seeking maximum force in the shortest amount of time
  - Benefits
    - Improved intermuscular coordination
    - Increased muscle size
    - Improved storage and use of elastic energy
    - Increased active muscle working range
    - Enhanced involuntary nervous reflexes
    - Increased muscular pre-activity
    - Enhanced motor coordination
- Example exercises
  - Split jumps
  - Speed skater
  - Cone drills

- Resistance training
  - Also called strength training; exercise with the explicit intent of increasing strength, endurance, muscle size (hypertrophy), or power
    - Muscular endurance: the ability of a muscle or group of muscles to continuously exert force against resistance over time
    - Hypertrophy: an increase in muscular size as an adaptation to exercise
    - Strength: the amount of force that can be created by a muscle or group of muscles
    - Power: the amount of force exerted by a muscle or group of muscles in a given amount of time
  - Strength categories
    - Relative strength: the individual's body weight in relation to the amount of resistance they can overcome
      - Calculation: 1RM / body weight = force per unit of body weight
    - Maximum strength: the ability of a muscle or muscle group to recruit and engage as many muscle fibers as possible
    - Starting strength: the ability to recruit as many motor units as possible instantaneously at the start of a movement.
    - Power: a combination of strength and speed; a muscle's ability to generate maximal tension as quickly as possible
    - Speed strength: the ability of a muscle or muscle group to absorb and transmit forces quickly
  - Benefits
    - Increased lean mass
    - Increased resting metabolic rate
    - Promotes bone development

- Additional considerations
  - Compound exercises engage multiple joints and muscle groups
    - They are typically completed before accessory exercises
  - Partial repetitions are those that are completed with an intentionally reduced range of motion

- Cardiorespiratory training
  - Cardiovascular fitness: the capacity to take in, transport, and utilize oxygen; specifically, the efficiency at which the heart and lungs can provide oxygen-rich blood to working muscle tissue
  - Aerobic is WITH oxygen; anaerobic is WITHOUT oxygen
  - Benefits
    - Reduces fatigue
    - Improves energy levels
    - Reduces depression
    - Reduces stress and anxiety
    - Prevents some types of cancer
    - Improves sleep
    - Improves mental acuity (sharpness of the mind, determined by memory, focus, concentration, and understanding)
- The cooldown
  - Offers both physiological and mental benefits
    - Allows the heart rate and body temperature to return to normal
    - Helps return the body to its resting state
    - Can help to alleviate DOMS (delayed onset muscle soreness)

Acute Training Variables - Chapter 9

The acute training variables are modified based on the client's abilities, desired training outcomes, and progress through their training program. The initial and ongoing client assessments provide the trainer with important information needed to select the appropriate variables for each individual. They include the following variables:

- Type: techniques, equipment, or methods used to complete an activity
  - Technique is just as important to client success as program design
  - Understanding how to properly progress or regress a client, in addition to choosing the right exercises, is important for trainers
    - Variables that can be manipulated to create a progression or regression include load (weight), tempo (speed), range of motion, movement complexity, or novelty
    - Progressions: modifications to acute training variables that increase the challenge of a movement pattern.
    - Regressions: modifications to acute training variables that decrease the challenge of a movement pattern.

- Exercise selection: specific exercises executed in a workout session
  - When selecting exercises, pick movements that target major muscle groups
- Exercise order: the order in which exercises are completed within a training session
- Intensity: measurable amount of force or effort given to an activity or exercise; often expressed as a percentage of effort compared to a person's maximum effort
  - One-repetition max (1RM): a single maximum-strength repetition with maximum load

TRAINING GOAL	INTENSITY (PERCENTAGE 1RM)
Muscular endurance	67 percent or less
Hypertrophy	67 percent–85 percent
Maximum strength	85 percent or greater
Power • Single-repetition event • Multiple-repetition event	 80 percent–90 percent 75 percent–85 percent

- Set: the number of times an exercise or group of exercises is completed

TRAINING GOAL	SETS
Muscular endurance	1–3 sets
Hypertrophy	3–4 sets
Maximum Strength	3–5 sets
Power	3–5 sets

- Repetitions (reps): the number of times an exercise is completed within a set
  - When deciding to increase load or repetition, the trainer should closely monitor responses to ensure clients are progressing toward their specific goal.

TRAINING GOAL	REPETITIONS
Muscular endurance	15 or more repetitions
Hypertrophy	6–12 repetitions
Maximum strength	1–6 repetitions
Power	1–5 repetitions

- Frequency: the number of times training occurs within a specific period or how often an exercise is executed
  - The client's overall level of fitness, as determined by assessments, is the most important factor to consider when prescribing exercise frequency
- Range of motion (ROM): the measurement of movement around a specific joint or body part
- Time: the duration of an activity or training session
- Tempo: the speed at which an exercise or movement pattern is completed
  - Tempo is written as:
    - eccentric count : isometric hold count : concentric count : isometric hold count
      - For example, completing a biceps curl with a tempo of three-count eccentric (lowering), no isometric hold, one-count concentric (lift or contract), and no isometric hold would read as 3:0:1:0
  - Closely related to time under tension (TUT)

TRAINING GOAL	TEMPO (EXAMPLE IN SECONDS)
Muscular endurance	4:0:6:0
Hypertrophy	3:1:3:1
Maximum Strength	3:0:1:0
Power	Fastest controllable tempo

- Time under tension (TUT): the amount of time a muscle is engaged as a set, completed from start to finish
- Volume: A measure of total work done
  - Sets x Reps x Intensity (Load)
- Rest: the amount of time spent in recovery between sets or repetitions

TRAINING GOAL	REST BETWEEN SETS
Muscular endurance	30–60 seconds
Hypertrophy	30–60 seconds
Maximum strength	2–5 minutes
Power	1–2 minutes

The most common fitness variables for cardiovascular training are those in the FITT Principle:

- Frequency of exercise
- Intensity of exercise
- Time or duration of exercise
- Type of exercise

The most common training variables for resistance training are the 5 R's:

- Repetitions: the number of times the range of motion is completed consecutively
- Rest: the time spent resting between repetitions or between sets
- Recovery: the time spent recovering between exercise sessions
  - Active recovery is low-intensity exercise that can promote and accelerate muscular and metabolic recovery
- Resistance: the load (weight) used for an exercise
- Range of motion: the total amount of joint movement used during an exercise

Program Design Principles - Chapter 9

The following training principles serve as the foundation for all fitness program development. These principles help fitness professionals apply the science of training to create safe and effective exercise programs.

- Principle of specificity: the concept that training must be specific to an individual's goals as the adaptations they will see will be based on the training completed
  - Assessments allow a trainer to uncover a client's starting point and plan the appropriate progression for their goals and abilities
- SAID principle: specific adaptations to imposed demands; stress on the human system, whether biomechanical or neurological, will require the body to adapt specifically to those demands
- Principle of variability: training programs must include variations in intensity, duration, volume, and other aspects of practice
- Principle of individual differences: there is no one specific way to train every client due to the uniqueness of each person
  - Fitness assessments provide information about a client's current fitness level and potential challenges; the data collected should be used to determine the appropriate training loads and progression
- Diminishing returns: everyone has a set genetic limit to their potential and, eventually, the effort put into training will no longer produce the same results
- Principle of progressive overload: the body must be forced to adapt to (or overcome) stress that is greater than what is normally encountered
- Principle of reversibility: clients lose the effects of training after they stop working out
- General adaptation syndrome (GAS): the three stages of adaptation the body goes through in response to stress
  - Alarm stage (shock): the body's initial response to stress; symptoms include fatigue, weakness, or soreness
    - Individuals will see gains in strength, but mostly through neuromuscular changes
    - This stage may last from two to three weeks

- Resistance stage (compensation): after around four to six weeks of continued exercise, the body will enter the resistance stage
  - Changes in the body include biochemical, mechanical, and structural
  - Clients may experience improvements in muscle size and strength
- Exhaustion stage (fatigue): may happen at any time in GAS
  - Symptoms in this stage mimic those of the alarm stage, but without adequate rest or recovery, the client may experience burnout, overtraining, injury, or illness
  - The fitness professional should conduct assessments and monitor the client for symptoms of overtraining
    - Overtraining: a condition that is a result of the accumulation of stress from both training and non-training activities resulting in long-term decreases in performance
      - Overtraining symptoms
        - Unusual muscle soreness
        - Delays in recovery
        - Performance plateaus or declines
        - Irritability
- Supercompensation Is the term that describes the post-training period when the trained function has a higher performance capacity than prior to the training period

- Periodization: an organized approach to training involving progressive cycling of various aspects of a training program during a specific time
  - Important to help clients stay injury-free and keep their bodies responsive to adaptations
  - When conceptualizing the development of a fitness program, periodization is the most commonly used term to refer to how a training program is broken down into discrete time periods
  - The time periods are referred to as macrocycles, mesocycles, and microcycles
    - Macrocycle: the overall training period, usually one year or more
    - Mesocycle: a training phase in the annual training plan; made up of three to nine microcycles
    - Microcycle: a one-week-long cycle of training sessions, or a single session
  - Undulating periodization: short durations of hypertrophy training alternated with short durations of strength and power training
  - Block periodization: highly concentrated, specialized workloads focused on achieving maximum adaptation
  - Linear periodization: progresses from low-intensity to high-intensity across the entire macrocycle
- Categories of training
  - Foundational training serves to prepare the body for movement and includes flexibility, mobility, balance, and core training.
  - Metabolic training is geared toward training both the aerobic and anaerobic energy systems.
  - Strength training serves to increase muscle mass, strength, endurance, and power.

TRAINING CATEGORY	FITNESS PROGRAM ELEMENT(S) AND TYPES
Foundational training	**Flexibility training** **Core training** **Balance training**
Strength training	**Resistance training**
Metabolic training	High-intensity interval training (HIIT) **Cardiorespiratory training**
Speed, agility, and quickness training	Reactive training Ballistic training Plyometric training Agility training

*Bolded program elements are usually part of most fitness programs, regardless of training goal.

Special Populations - Chapter 16, 17

Designing individualized training programs for every client is key to a personal trainer's long-term success. However, the needs of each client vary by individual, and those needs also change throughout the different stages of life. Therefore, it is essential for a fitness professional to understand the mental and physical differences of the various stages of life to effectively program and coach clients.

- Youth
  - Key considerations
    - Children are different from adults in that they have:
      - less blood volume
      - smaller airways and more soft tissue
      - faster respiratory rates and heart rates
      - more skin surface area in relation to weight
      - less muscle mass and fat mass
    - A child's vision doesn't fully develop until the age of eight
    - Children going through puberty have poor coordination and balance
    - Resistance and strength training are acceptable and, in most cases, desirable

- Benefits of exercise for youth
  - Higher self-concepts than those who are inactive
  - Has the potential to increase their self-efficacy, creativity, self-esteem, internal locus of control (a belief that the individual is responsible for their own success), test scores for cognitive functioning, and classroom behavior
  - 20–35% reduced risk of premature mortality
  - 15–65% less likely to have psychiatric (mental health) problems
  - Reduced risk of suicide
  - Increased muscle endurance and strength
  - Improved bone density
  - Reduced risk of physical injury
  - Improved self-esteem
  - Improved athletic performance
  - Improved heart and lung capacity and function
  - Reduced excess body fat
  - Reduced risk of depression
  - Reduced risk of general diseases (heart disease, cancer, type 2 diabetes, etc.)
  - Promotes healthy blood pressure
- Injury prevention
  - Always ensure proper instruction and form
  - Use proper training progression
  - Adjust exercise intensity to meet the child's physical capabilities
  - Always include a warm-up and cooldown
  - Always supervise children during the entirety of a training session
  - Use appropriate equipment and make sure the client knows how to use each piece correctly
  - Require proper attire and shoes for training
  - Ensure adequate rest during exercise sessions and promote rest on off days
  - Use cross-training to prevent injury and keep children engaged
  - Encourage optimal hydration and nutrition before, during, and after exercise
- Childhood development
  - Under 6
    - Growth slows after age 2
    - Sensitive period for motor skills development
    - Responsive to what they see, hear, and touch and will learn skills and movement patterns easily
    - Enjoys parallel play

        - Still learning to share, work in a team, and follow detailed instructions
        - Focus on gross motor skills
        - Ideal activities: freeze tag; duck, duck, goose; treasure hunts; T-ball; follow the leader; and kicking a ball into a goal
    - 6-10 years
        - Slow steady growth until adolescence
        - Ideal stage for both gross and fine motor skill development
        - Can successfully join and work in a group, manage conflict, take turns, find new friends, and manage relationships
        - High self-esteem and eager to learn new skills
        - Emphasize personal development and process goals over outcome goals
        - Ideal activities: bodyweight exercises, running and jogging, dancing, push-ups and pull-ups, sports activities, gymnastics, stretching, climbing, and wrestling
    - 11-19 years
        - Growth spurt
            - Girls: 10 to 13
            - Boys: 12 to 15
        - Dramatic hormonal changes
        - Often think and behave in their own best interest
        - Tend to think everyone is watching them
        - Can begin learning specialized motor skills, such as those pertaining to sports
        - Refine movement patterns
    - Youth fitness assessments

AREA OF FITNESS	PERFORMANCE ASSESSMENT
Aerobic capacity	PACER test, one-mile run, or walk test
Body composition	Skinfold, bioelectrical impedance (BIA), or body mass index (BMI)
Flexibility	Back-saver sit and reach, shoulder stretch
Muscular strength and endurance	Curl-up, trunk lift, 90-degree push-up, modified pull-up, flexed arm hang

- Older adults
    - Key terms
        - Senescence: the process or state of growing old
        - Chronological age: the number of years a person has lived
        - Functional capacity: the capability of performing tasks and activities that people find necessary or desirable in their lives

- Sarcopenia: the degenerative loss of skeletal muscle mass
- Dynapenia: the age-associated loss of muscle strength that is not caused by neurologic or muscular disease
- Metabolic syndrome: a cluster of at least three biochemical and physiological abnormalities associated with the development of cardiovascular disease and type 2 diabetes
- Frailty: an increased vulnerability resulting from age-associated decline in reserve and function across multiple physiologic systems
- Sleep apnea: a disorder of breathing during sleep
- Comorbidities: the simultaneous presence of two chronic diseases or conditions in a person
  - People with chronic health conditions often have more than one condition
- Parkinson's disease: a progressive disease of the nervous system marked by tremor, muscular rigidity, and slow, imprecise movement
- Osteoporosis: a skeletal condition that results in weak or brittle bones

- Aging affects the function of all body systems
  - Decreases in hormone production
  - Decreased elasticity
  - Diminishment in muscles, skin, and blood vessels
  - Recovery time is important with this age group
- Fitness professionals should consider the following when designing programs:
  - Between the ages of 20 and 80
    - Maximal oxygen uptake declines by approximately 50 percent
    - Maximal cardiac output declines by approximately 25 percent
    - Maximal heart rate declines by approximately 25 percent
    - Maximal stroke volume declines by approximately 15 percent
  - The average rate of decline (per decade) for adults over the age of 50 is as follows:
    - Loss of muscle mass declines by approximately 6 percent
    - Muscle strength declines by approximately 12 to 14 percent
    - Bone mass declines by approximately 10 to 15 percent

- Senior fitness tests

FITNESS TESTS	EVALUATE	HOW TO PERFORM	AT RISK
30-second chair stand	Lower-body strength for activities of daily living (ADL) such as getting out of a chair, gardening, or navigating stairs.	With arms folded across chest, count the number of stands that can be fully completed within 30 seconds.	Less than 8 unassisted stands for men and women
Arm curl	Upper-body strength that measures ADL such as carrying groceries, lifting grandchildren, or putting dishes away.	Count the number of biceps curls that can be completed within 30 seconds while holding a hand weight (5 lb. or 2.27 kg for women; 8 lb. or 3.63 kg for men).	Less than 11 curls for men and women
Two-minute step test	Stamina for activities such as shopping, traveling, or yard work.	Count the number of full steps completed within two minutes. Each knee must be raised to point midway between the kneecap and hip bone. Only the number of times the right knee reaches required height is scored.	Less than 65 steps for men and women
Chair sit and reach	Lower-body flexibility for preventing lower-back pain, balance, posture, and falls.	While seated at the front of a chair and with one leg extended, reach toward the toes with the hands. Count the number of inches between fingers and tip of toe.	Minus four inches or more for men Minus two inches for women
Eight foot up and go	Speed, agility, and balance to help in activities such as walking through crowds or playing at the playground with grandkids.	Count the number of seconds needed to get up from a seated position, then walk eight feet (2.44 m), turn, and return to a seated position.	Greater than 9 seconds for men and women
Back scratch test	Flexibility in the upper body, which may affect the ability to reach up high, change a light bulb, or open the refrigerator.	Begin standing. Place one hand behind the back and slowly move it up the spine toward the head. The opposite hand is placed behind the neck and is slowly moved down the spine with the goal of bringing both hands as close together (or overlapping) as possible. Repeat with the opposite hands (switch hand placement).	N/A

- Exercise program design
  - Cardiovascular guidelines
    - 150 minutes per week of moderate-intensity aerobic exercise (30-minute workouts five times per week)

        - 75 minutes per week of vigorous-intensity aerobic exercise (30-minute workouts three times per week)
        - Or any combination of the two
    - Resistance training
        - Start with closed kinetic chain exercises to support the joints
        - Safety and form are critical
        - Common modalities: bands, machines, free weights, and calisthenics
        - Guidelines
            - One set of 8–10 repetitions for exercises working the major muscle groups on two to three nonconsecutive days of the week
            - Ask the client to rate their exertion on a scale of 0 to 10, with 0 equating to no movement and 10 to an all-out effort
            - Moderate-intensity exercise (a perceived effort of 5 or 6) should allow the client to perform 12–15 repetitions and high-intensity exercise (a 7 or 8 on the scale) should permit between 8 and 10 repetitions
    - Flexibility training
        - Flexibility of the shoulder and hip joints decrease by approximately six degrees per decade between 55 and 86 years of age
        - Promotes optimal biomechanics and posture
        - Light static stretching of the major muscle groups at least twice per week can elicit improvements
        - Is best done after a thorough warm-up, after walking, or directly after a hot shower to improve muscle pliability
    - Balance training
        - Essential for seniors
        - Three days per week for about 10 to 15 minutes per session
        - Safety is of utmost importance
    - Mental training
        - Reducing sedentary behavior may improve brain health for those at risk for progressive mental degeneration conditions such as Alzheimer's disease
        - Both cognitive and physical training have positive effects on the brain
        - Moderate-intensity exercise can improve brain structure and function and may reverse neural decay
- Pregnancy
    - Key terms
        - Prenatal: occurring or existing before birth
        - Postpartum: the period of time following childbirth

- Relaxin: a sex hormone that facilitates birth by causing relaxation of the pelvic ligaments
- Progesterone: a female hormone that regulates the menstrual cycle and is crucial for pregnancy
- Hypermobility: the condition of having an excessive range of motion in a joint or joints
- Diastasis: the separation of the large abdominal muscles during pregnancy
- Gestational diabetes mellitus (GDM): a condition characterized by an elevated level of glucose in the blood during pregnancy, typically resolving after the birth

- Key physiological changes

BODY SYSTEM	OBSERVABLE CHANGES
Cardiovascular system	• Resting heart rate increases • There is a 45 percent increase in blood volume • Stroke volume and cardiac output increase • Respiratory rate may increase up to 50 percent • There is a 15–20 percent increase in oxygen consumption during exercise • Lung capacity decreases • There is a decrease in oxygen availability • Risk for orthostatic hypotension
Musculoskeletal system	• Bodyweight increases, placing more stress on joints • Elasticity of muscles, tendons, and ligaments increases • The center of gravity changes, causing issues with balance • Postural changes contribute to compensations
Endocrine system	• Levels of relaxin and progesterone increase • Thyroid-stimulating hormones released more often • Insulin resistance is increased due to a continuous supply needed for the fetus • More lipid (fat) use leading to ketogenesis

- Contraindications
  - Incompetent cervix
  - Persistent vaginal bleeding
  - Placenta previa after 26 weeks gestation
  - Premature labor
  - Preeclampsia
  - Uncontrolled diabetes or other systemic disorder

- Adaptive fitness
  - Key terms
    - Disability: a physical or mental condition that limits a person's movements, senses, or activities
    - Impairment: the state of being diminished, weakened, or damaged—especially mentally or physically
    - Activity limitation: the quantitative and qualitative measure of disability referring to difficulties experienced by an individual in executing a task or action
    - Participation restrictions: a problem experienced by an individual in involvement in life situations
    - Inclusion: the act of including into a group; encourages involvement and empowerment, where the inherent worth and dignity of all people are recognized
    - Adaptive physical fitness: the art and science of developing, implementing, and monitoring a carefully designed physical fitness program for a person with a disability
  - Those with disabilities do not have the same access to health care as others without disabilities

DISABILITY TYPE	PERCENTAGE OF POPULATION THAT DOESN'T GET EXERCISE	ESTIMATED NUMBER OF PEOPLE
Mobility Dysfunction	57 percent	34,770,000
Cognitive Disability	40 percent	24,400,000
Vision Disability	36 percent	21,960,000
Hearing Impairment	33 percent	20,130,000
No disability	26 percent	84,682,000

  - Fitness is for everyone and helps to:
    - Reduce the risk of heart disease, stroke, diabetes, and some cancers
    - Improve self-confidence and autonomy
    - Reduce feelings of depression
    - Increase quality of life and life satisfaction scores
    - Increase longevity
    - Reduce some symptoms of chronic medical conditions
  - Exercise program design
    - Fitness programming is fundamentally similar for clients with disabilities and those without
      - Some clients' disabilities may require alterations or subtractions in different components of fitness
    - All acute training variables can be manipulated for clients with disabilities
      - General recommendations for cardiovascular fitness
        - 30 minutes of moderate-intensity aerobic activity five times per week
        - 25 minutes of vigorous-intensity aerobic activity five times per week
        - Or any combination of the two

- Muscular strength and endurance
  - Moderate- to high-intensity exercises, moving all the major muscle groups on two or more days per week
- Competitive sports
  - Special Olympics
  - Paralympics
  - The American Association of Adapted Sports Programs (AAASP)

Chronic Conditions - Chapter 16

There are many chronic conditions a client may have or may develop over their lifetime. Several are related to lifestyle and eating patterns and can be addressed with modifications of the same. However, some chronic conditions are hereditary and require additional adjustments and considerations. A fitness professional must have a clear understanding of common chronic conditions they may encounter for several reasons: first, to identify the limitations a client may have; second, to better create an effective training program with these limitations in mind; and third, to identify if and when a client should be referred to a health care professional for further guidance or modifications to treat a chronic condition.

It is important for the trainer to understand that the information presented here does not qualify them to adequately serve demographics with chronic conditions. It is meant to provide foundational knowledge the trainer can then build upon. Fitness professionals who desire to serve populations with chronic conditions should seek out advanced training, stay within their scope of practice, and always refer to other professionals when appropriate. Personal trainers should also be aware of clinical practice guidelines and work to collaborate with a health care team as a supporting professional.

- Chronic disease: a condition lasting a year or more that limits daily activities and/or requires ongoing medical attention
  - Although a personal trainer should not treat or diagnose any medical condition, it's important to be familiar with common types you'll likely encounter
  - Hypertension: high blood pressure reading more than 140/90 mm Hg
    - Affects 75 million people
    - Risk factors
      - Obesity
      - Diabetes
      - Hereditary
      - Unhealthy diet
      - Physical inactivity
    - Exercise program design
      - Client should not expect to see changes in their blood pressure values until around week 12 in their training

- A doctor should be consulted before beginning with any client that has a blood pressure over 160/100 mm Hg
- A warm-up is essential (allows the blood vessels to slowly expand to accommodate greater blood flow)
  - It should last 10 minutes or longer
- No heavy overhead lifts
- The estimated maximum heart rate formula used for clients with high blood pressure is 162 – (0.7 x age)
- Vary exercises between the upper body and lower body
- Concurrent training is ideal; include both cardiorespiratory exercise and resistance training into a fitness program
- Beta-blockers (one of the most widely prescribed classes of drugs to treat hypertension) block specific receptors to reduce the effects of signals sent to increase heart rate
- The DASH diet (Dietary Approaches to Stop Hypertension) is ideal
  - Low-sodium, whole-food diet created for the treatment of hypertension

- Diabetes: a condition characterized by an elevated level of glucose in the blood
  - Key terms
    - Insulin: a hormone produced in the pancreas to regulate blood sugar
    - Prediabetes: a condition where blood glucose is higher than it should be, but not in the diabetes range
    - Type 2 diabetes: a long-term metabolic disorder characterized by high blood sugar, insulin resistance, and relative lack of insulin
      - The pancreas cannot keep up with the demand and begins to shut down
        - Insulin resistance: an impaired response of the body to insulin, increasing levels of blood glucose
      - An exercise stress test is highly recommended
    - Type 1 diabetes: the pancreas produces little or no insulin
      - Diet and lifestyle habits do not cause this condition
      - Is an autoimmune condition
      - Need to take insulin shots (or wear an insulin pump) every day
    - Gestational diabetes: an elevated level of glucose in the blood during pregnancy, typically resolving after birth
  - Risk factors
    - Hyperglycemia: elevated blood glucose
    - Overweight
    - Age; risk increases after age 45

        - Family history
        - Ethnicity—Black, Hispanic, American Indian, and Asian American people are at a higher risk for prediabetes and type 2 diabetes
    - Exercise program design
        - Low- to moderate-intensity activity is safe
        - Moderate- to high-intensity training has been shown to reduce the risk of death for clients with diabetes
- Arthritis: pain or stiffness in skeletal joints (inflammation of the joints)
    - Key terms
        - Osteoarthritis: degeneration of joint cartilage and the underlying bone
            - Possible symptoms: pain and stiffness, fatigue, anxiety, depression, headaches, and memory problems
        - Rheumatoid arthritis (RA): a progressive disease causing inflammation in the joints
        - Fibromyalgia (FM): characterized by widespread musculoskeletal pain, fatigue, and tenderness in localized areas
        - Abnormal pain perception processing: an increase in the subjective interpretation of discomfort due to abnormal sensory processing in the central nervous system
        - Gout: a disease in which defective metabolism of uric acid causes arthritis
        - Flare: a sudden surge in rheumatoid arthritis inflammation
        - Remission: a significant reduction in symptoms and signs of rheumatoid arthritis
    - Risk Factors
        - Age; risk increases with age
        - Sex; women more prone
        - Obesity
        - Genetics
        - Overuse
        - Smoking
        - Health history
            - Women who have never given birth
            - Lupus: a chronic autoimmune disease that creates inflammation and pain in various parts of the body
            - Post-traumatic stress disorder (PTSD): a persistent mental and emotional stress that occurs as a result of injury or psychological shock
        - Diet
    - Exercise program design
        - Warm-up and slowly move joints through their full range of motion
        - Isometric exercises
        - Start with as little as 5 or 10 minutes of physical activity, two or three times per day, until they can reach the following recommendations:

        - 150 minutes of moderate-intensity aerobic activity
        - 75 minutes of vigorous-intensity activity
        - Or a combination of moderate- and vigorous-intensity exercise (one minute of vigorous activity is similar to two minutes of moderate-intensity activity)
        - Include low-impact exercises such as brisk walking, cycling, swimming, water aerobics, light gardening, group exercise classes, or aerobic dance classes
    - Incorporate flexibility training for 10-15 minutes per session
    - Work out when pain and discomfort are minimal and joints are flexible
    - Avoid taking pain medications before an exercise session
    - 48 hours rest between sessions
    - Suggested resistance training progressions:
        - Week 1: one set of 15 repetitions at 60 percent 1RM
        - Week 2: two sets of 15 repetitions at 60 percent 1RM
        - Week 3: three sets of 15 repetitions at the same intensity
        - Weeks 4–6: three sets of 12 repetitions, increasing to 70 percent 1RM
        - Weeks 7–24: three sets of 8 repetitions, progressing to 80 percent 1RM
  - The CDC suggests following a SMART plan of action
    - S: start low and go slow
    - M: modify activity when symptoms increase but try to stay active
    - A: activities should be “joint-friendly”
    - R: recognize safe places and ways to be active
    - T: talk to a health professional or certified exercise specialist
- Coronary heart disease (CHD): the narrowing and even blockage of the arteries and blood vessels
  - Can sometimes go without symptoms, but the number one sign of this condition is anginal chest pain
  - Every 40 seconds, someone in the United States has a heart attack
    - Signs and symptoms of a heart attack include:
        - Chest pain or discomfort
        - Upper-body pain, such as in the arms, back, neck, jaw, or upper stomach
        - Shortness of breath
        - Nausea, lightheadedness, or cold sweats
  - Risk factors
    - Obesity
    - Physical inactivity
    - High blood pressure
    - Smoking
    - Diabetes

- Exercise program design
  - Aerobic exercise intensity should gradually increase until the client can participate in the following recommendations:
    - 150 minutes of moderate-intensity aerobic activity per week
    - 75 minutes of vigorous-intensity aerobic activity per week
    - Or any combination of the two
  - Get medical clearance from a physician
  - High impact activities will not be suitable for this population during the early stages of programming and heart rate should be carefully monitored

- Asthma: a respiratory condition marked by spasms in the bronchi of the lungs, causing difficulty in breathing
  - Environmental considerations are key for this population and include pollen, allergies, smoke, extreme temperatures, and chemicals
  - Key terms
    - Exercise-induced bronchoconstriction (EIB): asthma attack triggered by doing sports or physical activity
    - Gastroesophageal reflux disease (GERD): a condition in which acidic gastric fluid flows backward into the esophagus, resulting in heartburn
    - Chronic obstructive pulmonary disease (COPD): a lung disease characterized by chronic obstruction of lung airflow that interferes with normal breathing and is not fully reversible
    - Hyperventilation: to breathe at an abnormally rapid rate, increasing the rate of loss of carbon dioxide
    - Wheezing: breathing with a whistling or rattling sound in the chest
    - Corticosteroids: a group of natural and synthetic steroid hormones produced by the pituitary gland
  - Risk factors
    - Age; more prominent in children
    - Sex; occurs more often in boys than girls
    - Family history
    - Allergies (such as to dust mites, mold, and pollen)
    - Smoking
    - Obesity
      - Obese clients are 92 percent more likely to develop asthma compared to the average client
  - Exercise program design
    - Training environment should be free from allergens and pollutants that could trigger the client's asthma
    - Initial recommendation for intensity of aerobic training is 50 to 60 percent of VO_2 max
    - Should not exercise in excess of 80 percent VO_2 max